

Adapting and Simplifying Systems Engineering Frameworks to Support Start-Ups with Limited Resources in Transitioning from Prototypes to Commercial Products

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Abstract. This research lays the groundwork for a future master's thesis, focusing on the adoption and modification of a lightweight Systems Engineering (SE) framework specifically designed for start-ups and small companies operating in complex industries. These organizations often face unique challenges, including limited resources, fragmented stakeholder involvement, and difficulties in transitioning from traditional engineering models to more suitable SE frameworks. To address these challenges, the study examines lighter alternatives such as the Model-Based Systems Engineering (MBSE) framework and the CAFCR model.

The study uses a Norwegian start-up as a case study, focusing on its efforts to transform its prototype smart sorting bin system into a commercially viable product capable of managing various types of solid waste. By examining the company's operational context and unique challenges, the research aims to identify how a tailored SE framework can help address its specific needs and enhance its processes.

Keywords. Adoption of Systems Engineering (SE), Lightweight SE Framework, Limited Resource Enterprises, Roadmapping for Master's Thesis, Start-Ups, Small Enterprises, Smart Sorting Bin System, Methodological Guidelines.

Introduction

The main objective of this article is to outline a research roadmap for a master's project and thesis that explores the application and modification of SE frameworks in start-up companies with limited resources. Traditionally, Systems Engineering techniques have been applied in large industries, such as aerospace and defense, where systems are complex and resources are abundant (Muller, 2009; Younse et al., 2021). However, start-ups and small enterprises often operate under very different conditions, facing unique challenges such as resource constraints, limited stakeholder engagement, and the need for rapid innovation (Muller, 2009; Nanfuka & Oosthuizen, 2023). This article aims to explore the existing academic knowledge of how a simplified, lightweight version of the SE framework can be adapted to assist these companies in transitioning from prototypes to commercially viable products for application in the master's project and thesis.

The article examines existing literature discussing a modified SE framework that fits the needs of start-ups and small enterprises. The case study of a Norwegian start-up, which is working to develop its prototype of a smart sorting bin system into a commercially scalable product, will help assess the practical challenges faced by such companies and how a streamlined SE framework can be effectively applied to meet their needs.

This article also outlines the research methodology that will be used in the master's project and thesis to explore the possibilities of modifying and adopting the SE framework in the context of smaller, innovation-driven companies by focusing on the early stages of product development, stakeholder alignment, and resource-efficient methodologies for helping start-ups overcome the challenges they face in competitive and fast-moving industries.

The context

The scope. This research investigates the application and adaptation of a lightweight (SE) framework for a Norwegian start-up working to commercialize its prototype of a smart waste-sorting system. The research explores how a modified SE framework can be tailored to meet the unique challenges faced by start-ups, including the need for rapid innovation, resource efficiency, and stakeholder alignment across diverse markets. The research focuses on optimizing the SE framework to maximize its benefits while minimizing resource use and development time, addressing stakeholder needs, and ensuring the system performs consistently across different operational environments.

The company. Infinity Fabrtkk AS is a Norwegian startup established in 2022 to develop solutions aimed at increasing the percentage of private solid waste sorting in Norway, the Middle East specifically, and globally in general. The company has gained experience through the development of a smart sorting machine designed for plastic bottles and metal cans, which has been implemented and executed in the Middle East. Currently, the company is working on designing and producing smart sorting machines capable of handling the five main types of solid waste: glass, metal, paper, organic matter, and plastic.

The development of this machine is partially funded by Innovation Norway. By September 2024, the company achieved a functional prototype that validates the concept. However, due to several factors, such as the short development timeline and limited resources—both human and financial—the company was unable to apply the systems engineering approach during the prototype development phase.

However, after the completion of the prototype, several investors expressed interest in supporting the company upon the achievement of its first commercial version of the machine. These investors come from two distinct markets: the Norwegian market and the Middle East, each with unique key drivers and requirements.

The company now aims to apply a systems engineering approach to develop the first commercial version of the machine that fulfils the needs and the key drivers of stakeholders in both markets. However, the company faces constraints, including a short delivery timeline, set for the second quarter of 2025, and limited financial resources which create the need to adopt an optimized systems engineering framework to streamline the development process within a short time frame and with minimal resources.

The System. The solution presented by Infinity Fabrtkk AS is a system designed to control and supervise the accuracy of a user's pre-sorting of waste before it is inserted into the system. Based on the accuracy and purity of the pre-sorted waste, the system determines whether to allow the waste to enter the designated material-specific waste basket or to reject it, directing it instead to the unsorted basket.

The system also offers rewards to encourage users to engage with it and provide pre-sorted waste with high accuracy. On the other hand, in the case of low waste-sorting accuracy, a portion of the rewards is withheld until users participate in environmental awareness courses specifically tailored to address the types of errors they made, thereby promoting the user sorting accuracy in the future.

Additionally, the system collects and analyzes statistical data to create a comprehensive big data that generates valuable information, which can be used by governments, organizations, and institutions for research related to environmental awareness.

Furthermore, the system monitors the waste levels in each basket and analyzes usage patterns to create models and plans that assist waste collection companies in optimizing their operations. This ensures waste is collected in the most efficient ways, both economically and environmentally, contributing to a reduction in carbon emissions during the process.

the problem statement

During the prototype development process, the focus of the engineering team was on identifying a technical solution to solve the gap in the market based on the minimum technical requirements and the knowledge gained from their previous projects. However, a deep analysis of the context and environment in which the system would operate was not conducted. Moreover, the analysis of stakeholder needs primarily focused on the most directly involved stakeholders, overlooking the various other stakeholders and their requirements across the different target markets.

When the company began the development process for the first commercial version of the system, it was observed that several stakeholders with diverse needs had not been considered, creating a risk of developing a functional system that does not fully satisfy stakeholder requirements. Furthermore, significant variations were identified between the context and operating environments of the target markets, which could result in a system with inconsistent performance across different operational environments.

Given the current situation, the company faces constraints of limited time and resources and must adopt an optimized systems engineering framework for the development of the first commercial version of the system. This framework needs to maximize the benefits of the systems engineering methodology while avoiding overburdening the company's limited resources and preventing delays in the system delivery schedule.

The Research Objectives

The objectives of this research can be categorized into one main objective and two sub-objectives, as outlined below.

The main Research Objective. To find and develop an optimized, lightweight method for applying systems engineering frameworks that effectively meets the needs of startups with limited resources and tight delivery timelines.

The Research Sub-Objective.

To assess which are the most critical stages of the system engineering framework model for startups with limited resources and tight delivery timelines.

To assess which is the most optimized system engineering model that fits the needs of the startups with limited resources and tight delivery timelines.

The Research Questions

Based on the research objectives and sub-objectives, it becomes possible to outline the main research questions and sub-questions that the study aims to address.

The main Research Question. How can a simplified and optimized method for applying systems engineering frameworks be developed to effectively meet the needs of startups with limited resources and tight delivery timelines?

The Research Sub-Questions.

What are the key challenges startups face when applying traditional engineering methods with limited resources and tight delivery timelines?

Which stages of the systems engineering framework are most critical for startups to focus on, considering resource and time constraints?

Which systems engineering model is the most optimized for the needs of startups with limited resources and tight delivery timelines, ensuring successful project delivery?

The Literature Review

A comprehensive review of relevant literature was carried out to build the necessary academic foundation for this research. The process followed the guidelines provided in *the Systems Engineering Course Research Method*(Muller, 2024)s. The research objectives and questions informed the development of various keyword combinations, which were applied across three academic search engines: Oria, Gaudí System Architecting, and Google Scholar. Published articles were identified, and multiple layers of filtering were applied to streamline the results, focusing on the most relevant studies.

The research keyword combinations were based on terms related to *Systems Engineering Frameworks, Startups and Small Companies, Adopting Systems Engineering, and the Challenges of Using Traditional Engineering Models*. The following diagram illustrates the most important results and the filtration process that was applied.

As shown in the diagram below , the literature evaluation and filtration processes resulted in 18 articles identified as relevant to the research topic and questions. However, 10 of these articles were selected to serve as the primary foundation for the research, while the remaining articles were designated as secondary resources. The reasons for this selection are described in the detailed review of the articles below.

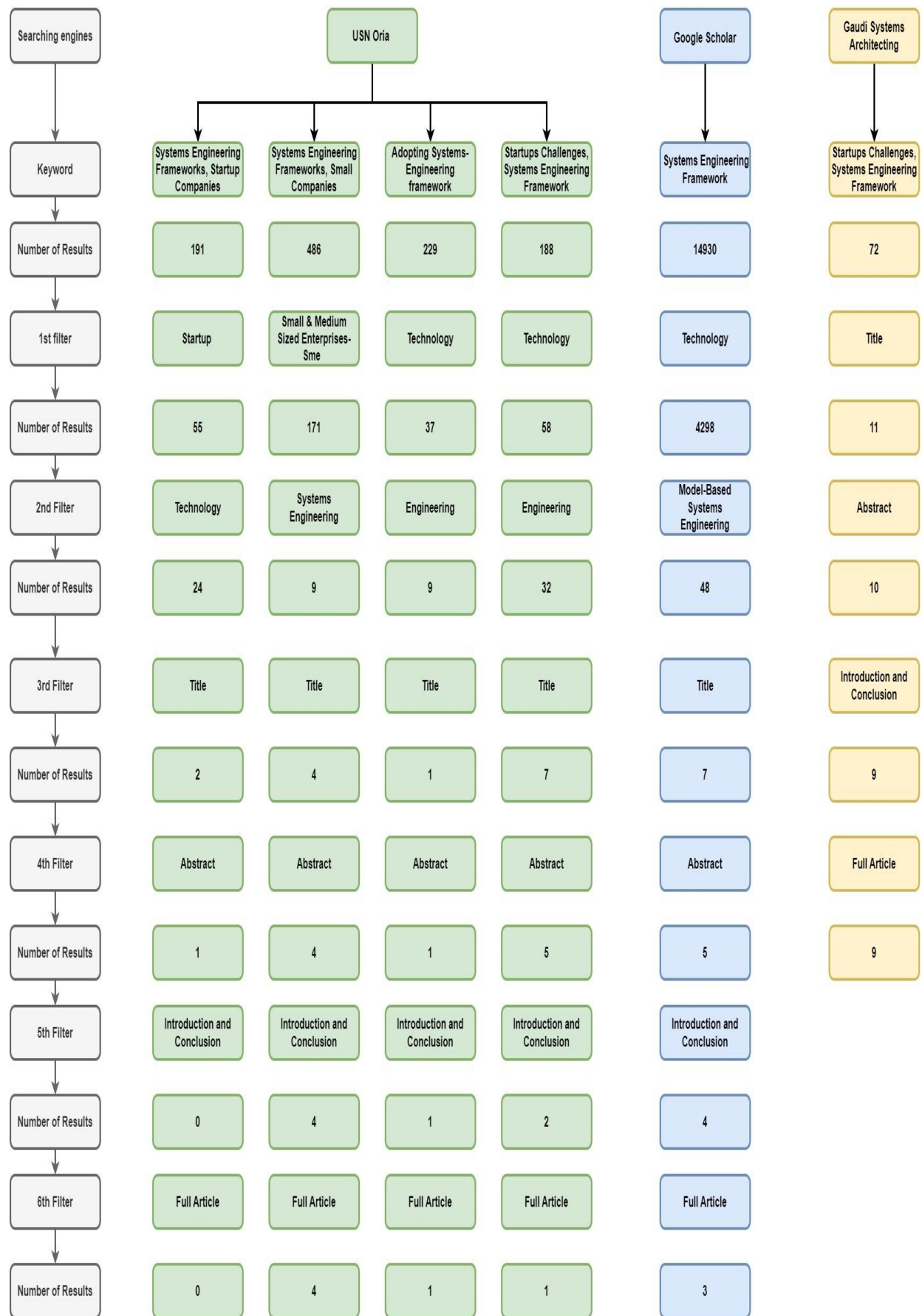


Figure 1- Literature filter results

The first article, titled *MBSE-LITE: A Framework for Adopting Model-Based Systems Engineering in Small and Medium-Sized Enterprises in South Africa*, authored by Justine Geraldine Nanfuka and Rudolph Oosthuizen, is considered one of the most relevant papers to the research topic. This article investigates the factors influencing the adoption of the systems engineering framework among small companies, with the participation of 100 businesses in South Africa.(Nanfuka & Oosthuizen, 2023)

The article discusses the historical background of the founding of systems engineering and how the development of this field was primarily focused on large companies, with little attention given to the needs of small companies that is normally facing several challenges as the limited financial resources, human resources, and training opportunities.(Nanfuka & Oosthuizen, 2023)

Model-Based Systems Engineering (MBSE) is discussed in the article as a modification of the systems engineering framework that may address some of the challenges faced by small companies. To provide a clearer understanding of MBSE, its definition as presented by the International Council on Systems Engineering (INCOSE) is highlighted:

“Model-based systems engineering (MBSE) is the formalized application of modeling to support system requirements, design, analysis, verification, and validation activities beginning in the conceptual design phase and continuing throughout development and later life cycle phases.”(Hart, 2015; INCOSE, 2007)

The researchers delved deeper into the analysis of transitioning from the Systems Engineering (SE) framework to Model-Based Systems Engineering (MBSE). The results of this analysis were found to be particularly relevant to the research topic, as 80% of the sample successfully transitioned to MBSE. This suggests that the risk associated with using a liter model is not high, especially when considering the causes identified in the article for the 20% of participants who failed to transition to MBSE. These causes are listed in the article.(Nanfuka & Oosthuizen, 2023)

Another interesting aspect of this article is that it introduces a modified and lighter version of Model-Based Systems Engineering (Lite-MBSE), which aligns with the goals of this research. In this study, a modified model will be developed that is more suitable for small and emerging companies, similar to the approach presented in the paper.(Nanfuka & Oosthuizen, 2023)

Additionally, the article provides valuable recommendations on how small companies can successfully adopt Lite-MBSE, based on the authors’ observations during their research. These insights will be crucial when preparing recommendations for the target company in this study. However, it is important to consider the differences in context, as this research focuses on start-ups operating in a highly competitive innovation environment, which may require tailored strategies compared to those suggested in the paper.(Nanfuka & Oosthuizen, 2023)

Another article that focuses on Model-Based Systems Engineering (MBSE) is titled *Comparative Analysis of a Model-Based Systems Engineering Approach to a Traditional Systems Engineering Approach for Architecting a Robotic Space System Through Knowledge Categorization*, authored by Paulo J. Younse, Jessica E. Cameron, and Thomas H. Bradley. In this study, a robotic space system is used as a case study to compare the Systems Engineering (SE) framework and MBSE across different stages of the project life cycle.(Younse et al., 2021)

This comparison is important and relevant to this research as it provides a deeper understanding of the differences between the two approaches and helps identify where and how the SE framework can be modified without compromising its efficiency using one of the most complex systems with the lowest risk tolerance as an example. Therefore, this article offers a valuable perspective on adapting the SE framework to fit the needs of the startups and the small companies, which aligns with the objectives of this study.

The adoption of the SE framework in industries outside the aerospace and defence sectors is recognized as a significant challenge, as many of these industries rely on traditional engineering models. However, the growing complexity of the projects undertaken in such industries necessitates the adoption of more suitable frameworks to leverage the added value that models like the SE framework can provide. To address these challenges, an article titled *Lessons Learned from the Introduction of Systems Engineering*, authored by Daria, Robin, Lukas, Harald, Anac, and Roman, serves as a foundational resource for this research.(Wilke et al., 2023)

This article explores the challenges associated with adopting the SE framework in non-aerospace industries, examining the factors that contribute to hesitation or resistance among companies. It also highlights key motivators and enablers that encourage organizations to implement the framework despite these challenges.

A central contribution of the article is the presentation of 78 hypotheses and guidelines related to the most efficient practices of the SE, of which 40 have been quantitatively and qualitatively verified. These verified recommendations offer valuable insights and should be carefully considered when developing guidance for the target company in this research.(Wilke et al., 2023)

By incorporating these findings, the modified SE framework proposed in this study can be tailored to meet the specific needs and address the concerns of the company. This approach aims to maximize the framework's effectiveness and ensure its acceptance, providing a pathway to successfully adopt and benefit from SE practices in a broader range of industries.

As described in the context section, the case study of this research focuses on a company engaged in an innovation project with limited analysis of the stakeholders involved. For this reason, the article titled *Early Validation of User Needs in Concept Development: A Case Study in an Innovation-Oriented Consultancy*, authored by Kjørstad, Falk, Muller, and Pinto, has been selected as a primary resource for this study.(Kjørstad et al., 2019)

The article explores the importance of understanding user needs—representing one of the most critical stakeholders—early in the systems engineering (SE) framework, particularly from an innovation-oriented perspective. Also, it highlights how understanding stakeholder requirements early on can greatly reduce the risk of expensive design changes later in the process. This point is particularly relevant to this research, as the case study involves limited analysis of the stakeholders involved.(Kjørstad et al., 2019)

The article also argues that human factors, particularly the needs of the user as a key stakeholder, often receive less attention compared to stakeholders with technical perspectives. This imbalance is identified as a critical gap in SE practices that must be addressed to improve the overall framework's effectiveness.(Kjørstad et al., 2019)

Additionally, the article discusses several methods for early validation of user needs, such as Concept of Operations (ConOps), storytelling, and rapid prototyping. These methods are analyzed in terms of their value in ensuring a thorough understanding of user needs during the early stages of the SE framework. Such insights provide practical and theoretical guidance that

is integral to shaping the modified SE framework proposed in this research, ensuring a balanced consideration of technical and human factors.(Kjørstad et al., 2019)

Another article that focuses on a similar subject as the article above is titled *Early Validation of Stakeholder Needs by Applying Co-creation Sessions*, authored by Guntveit, Kjørstad, and Kjørstad. This article explores the added value that innovation-oriented consultations bring in capturing the needs of key stakeholders at the early stages of a project. It emphasizes how these consultations help to integrate stakeholder needs effectively into the project outline and early design stages. By doing so, they create a shared understanding between stakeholders and the implementing companies, ensuring that the stakeholders' requirements are thoroughly understood and addressed.(Guntveit et al., 2020)

The article also provides an in-depth explanation of various approaches to conducting co-creation sessions, examining their effectiveness in achieving these goals. Different methods are analyzed and tested, with the results offering valuable insights into how early stakeholder involvement can be optimized to benefit the overall project outcomes. This analysis underscores the importance of such sessions in aligning project objectives with stakeholder expectations, contributing significantly to the project's success.(Guntveit et al., 2020)

An essential perspective for this research is presented in one of the most significant articles it relies upon, titled *Systems Engineering and Modeling at Start-Up Company*, written by Gerrit Muller. The article examines the use of (SE) in a start-up company. The article takes a start-up operating in a complex industry—copper structure printing—as a case study, providing valuable insights into the application of SE in startups dealing with complex systems.(Muller, 2009)

Additionally, the work presented in this article highlights, in one way or another, the most important stages within the SE framework for start-ups as system modeling, contextual understanding, workflow design, and process flow are thoroughly explored. These stages are essential to consider when modifying the SE framework in this research and need to be taken into account during the adaptation process, ensuring that the modified framework effectively addresses the unique challenges faced by start-ups.(Muller, 2009)

Furthermore, the article concludes that applying Systems Engineering facilitated early verification at the system level. This early validation played a significant role in reducing the number of risks identified during the process, ensuring they were addressed effectively. This conclusion underscores the practical benefits of using the SE framework in start-ups, particularly in complex and high-risk industries, making it a cornerstone resource for this research.(Muller, 2009)

Another important source for this research is a thesis titled *CAFCR: A Multi-view Method for Embedded Systems Architecting; Balancing Genericity and Specificity*, written by Gerrit Muller. In addition to its thorough explanation of various aspects of systems engineering, this thesis introduces an interesting model called CAFCR. The CAFCR model divides system design into five perspectives: Customer Objectives, Application, Function, Conceptual, and Realization. These perspectives aim to answer the most critical questions of any project—what, why, and how—while also considering both customer and product viewpoints.(Muller, 2004)

The thesis further explores several early validation methods, including Storytelling and Threads of Reasoning. These methods and the CAFCR model are examined by applying them to complex project, such as the Medical Imaging MRI, which is sourced from one of the most

intricate industries with a zero-tolerance culture. This case study serves as a key reference, providing valuable insights into how early validation techniques can be applied in complex, high-risk projects, and provides an important model as CAFCR that will be taken into consideration in this research.(Muller, 2004)

Another significant method within the SE framework is roadmapping, which is discussed in an article titled *Model-Based User Experience-Focused Roadmapping*, authored by Yuskevich, Hein, Doufene, and Jankovic. This article explores the traditional approach to creating roadmaps and introduces an innovative method referred to as model-based technology roadmapping. The proposed method focuses on integrating model-based techniques to improve the clarity and utility of the roadmapping process.(Yuskevich et al., 2024)

The article also presents the development of a unified language that can be utilized across various perspectives, including market analysis, UX design, engineering design, and technology. This unified language aims to streamline communication and collaboration among diverse stakeholders involved in the roadmapping process.(Yuskevich et al., 2024)

Furthermore, the article highlights a recurring issue observed in the SE framework: the lack of sufficient focus on the user as a key stakeholder during the roadmapping process. This observation aligns with the findings of previously discussed articles regarding the underrepresentation of user needs in stakeholder analyses.(Yuskevich et al., 2024)

A notable conclusion of the article is that incorporating the proposed model-based approach with SysML modeling significantly simplifies the roadmapping process. This integration makes the process more straightforward, easier to learn, and more consistent, offering a practical solution for addressing some of the challenges associated with traditional roadmapping methods.(Yuskevich et al., 2024)

The final significant approach examined in this research is the use of the Concept of Operations (ConOps). To explore this approach, two articles discussing its applications have been selected. The first article, titled *Evaluation of Visual ConOps in Early Solution Validation in a Small and Medium-Sized Enterprise*, authored by Hadadpour, Kjørstad, and Muller, provides valuable insights into the use of ConOps for the early validation of solutions within small companies.(Hadadpour et al., 2024)

This article emphasizes the importance of employing ConOps to mitigate the risks associated with misunderstandings between internal stakeholders within the company and external stakeholders, such as clients and partners. By bridging these gaps, the method significantly reduces the likelihood of misinterpreting stakeholder needs and key drivers, which are critical for project success.(Hadadpour et al., 2024)

The study further highlights that the application of ConOps enhances engineers' understanding of the real needs and expectations of stakeholders. Importantly, this improvement is achieved without imposing additional strain on the already limited resources of small companies. The conclusion underscores the practicality and effectiveness of using ConOps in resource-constrained environments, making it a highly relevant approach for the context of this research.(Hadadpour et al., 2024)

The final article that this research draws upon is titled *Systems Engineering Research Methods*, authored by Gerrit Muller. This article serves as a foundational resource by offering a comprehensive overview of scientific research methods tailored to the field of systems engineering. Its insights are particularly valuable for ensuring that this research achieves both academic and

practical standards, providing a robust framework for conducting the study effectively.(Muller, 2013)

The article acknowledges that systems engineers often possess more practical, hands-on experience than academic or research expertise. Therefore, it provides a concise yet thorough explanation of various research tools and methodologies, designed to accommodate the time constraints typically associated with master's theses. By doing so, it ensures that these tools are accessible and practical for engineers transitioning into academic research.(Muller, 2013)

Furthermore, the article offers detailed guidelines for maintaining the quality and efficiency of the research process. These guidelines ensure that the methods used are both rigorous and efficient. This resource plays a critical role in shaping the methodological approach of this research, aligning it with the best practices in the field of systems engineering while addressing the unique challenges faced by researchers with practical engineering backgrounds.(Muller, 2013)

THE Research Methodology

To address the research questions, a combination of theoretical and practical steps will be implemented to analyze the primary challenges startups face when using traditional engineering methods. The focus will also be on identifying the most critical stages of the systems engineering process that startups need to adopt to overcome these challenges without overextending their limited resources and tight delivery timelines.(Muller, 2013, 2024) The methodology will involve the following steps:

A series of both quick and in-depth interviews will be conducted with the company's engineers across various fields to identify the challenges and knowledge gaps they face at this stage of development. This will help in understanding the critical needs in the process of transforming the prototype into a commercial product.

Additionally, project stakeholders will be analyzed, and interviews will be held with representatives from different stakeholder groups. These interviews aim to gather insights into their needs and expectations for the final product and compare these with the current expectations of the project outputs to identify gaps or areas of misalignment.

The information gathered from these interviews will then be analyzed collaboratively with the company's engineers and stakeholders. This analysis will help identify gaps between the current process outcomes and the stakeholders' needs, highlighting the weaknesses in the engineering methods currently used. Based on these insights, the study will determine which stages of systems engineering models should be integrated to address these weaknesses.

To build a strong theoretical foundation, a review of literature and scientific articles on applying systems engineering frameworks in startups and small companies will be conducted. This review will provide a broader understanding of how these frameworks can be implemented effectively without overwhelming the limited resources of such companies.

Each stage of the systems engineering framework will also be evaluated on an academic basis to assess its benefits for small companies. The analysis will focus on determining the necessity of each stage, its value to the company, and the resources required to integrate it into existing processes.

The most widely used systems engineering models will be reviewed and compared based on their benefits and the resources needed for their implementation. This comparison will aim to identify the most suitable model for startups, considering their unique circumstances and critical needs.

Based on the findings from the previous steps, recommendations will be developed for the integration of an optimized version of the systems engineering framework into the company's operational model. The implementation of this model will be observed, and its impact on the company's processes will be monitored, with additional data collected to refine the findings.

Following implementation, further interviews will be conducted with the company's engineers to gather their feedback and insights. These interviews will evaluate the impact of the systems engineering model on achieving desired results, the ease of its integration into workflows, and its influence on collaboration across departments.

Finally, interviews will also be conducted with stakeholders after the project's completion to evaluate their satisfaction with the delivered product. These discussions will assess how well the final outcomes meet their needs and expectations.

The findings from all these steps will be synthesized, allowing conclusions to be drawn and recommendations to be generalized regarding the impact of a lightweight systems engineering model in solving the challenges of limited resources and tight timelines faced by startups and small companies.

The Discussion

While the project objectives are clear, the research faces several significant challenges and risks. One of the most prominent concerns is that the scope of the research may prove too broad to be effectively explored within the short time frame available, which is limited to one semester. This constraint requires careful consideration of which aspects of the SE framework to modify and test, ensuring a focused and achievable approach.

The research also encounters challenges due to a lack of previous academic work on modifying SE frameworks specifically for startups and small enterprises. While SE is well-established in large industries such as aerospace and defense, there is limited literature on how these frameworks can be adapted for smaller, resource-constrained organizations. (Muller, 2009; Younse et al., 2021) This gap necessitates further exploration and will likely require the research team to conduct original investigations and compile relevant data. The scarcity of published articles that address the adoption and modification of SE frameworks in startup environments means that much of the research will be exploratory, contributing new knowledge to the field.

Another challenge lies in the complexity of applying the modified SE framework to a case study that involves two distinct markets—Norway and the Middle East. Each market comes with its own unique set of stakeholder requirements, business practices, and environmental conditions. This dual-market focus adds a layer of complexity to the research, potentially increasing the amount of time and resources needed for the study. While this challenge is significant, it also offers an opportunity to test the framework's adaptability and scalability across different contexts. By refining the model to work effectively in both markets, the study can provide a more robust solution that is applicable in diverse operational environments.

Additionally, the startup company's lack of prior experience with SE practices presents a risk to the research. The application of a modified SE framework in an organization unfamiliar with such methodologies could lead to errors or misinterpretations that might skew the results. To mitigate this risk, the research will require close supervision, continuous feedback, and iterative adjustments to ensure the model is being applied correctly and yielding accurate results. Continuous monitoring will also help identify any challenges or obstacles during the implementation process, allowing for timely adjustments to keep the research on track.

Another key concern is the involvement of both internal and external stakeholders throughout the research process. The success of this project depends not only on the startup's ability to implement the modified SE framework but also on the cooperation of stakeholders who must provide valuable insights and feedback. The research will need to establish clear communication channels, set expectations for stakeholder involvement, and develop strategies for effective data collection. Stakeholder engagement is critical, and ensuring their active participation will help align the framework with the company's objectives and the broader market requirements.

Despite these challenges, this article has laid the groundwork for a clear and structured approach to understanding and addressing the unique difficulties faced by startups in implementing SE frameworks. The roadmap established in this paper offers strategies to tackle these challenges, adapt the SE framework to smaller organizations for provide dynamic solutions that cater to the fast-paced, resource-constrained environments typical of startups.

The Conclusion

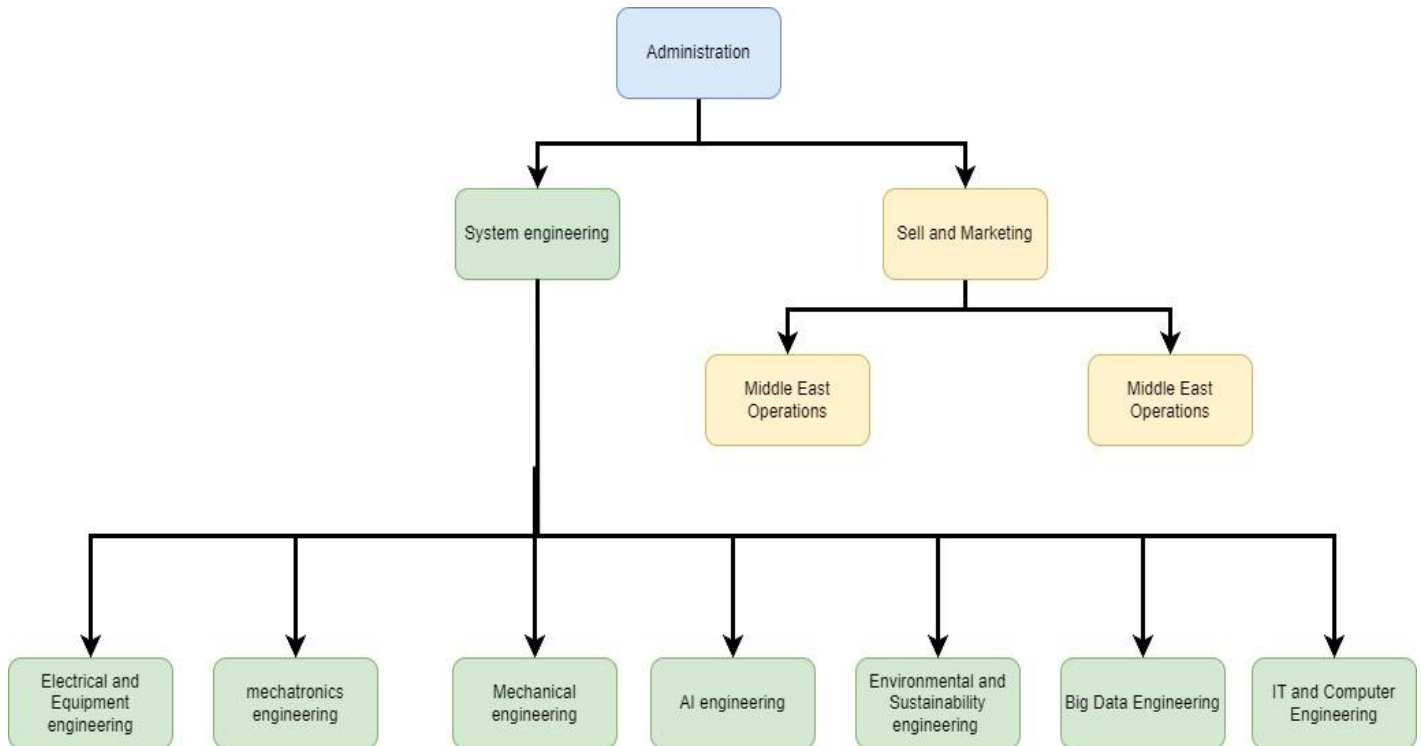
This article sets the stage for a master's project focused on exploring the modification of the Systems Engineering (SE) framework to better suit the specific needs of startups and small companies operating under constraints such as limited resources and time. The literature that will be used to gain academic knowledge and the research methodology that will be employed to implement the master's project are also highlighted.

The main objective of this research is to simplify and adapt the SE framework, making it more accessible for companies with limited experience in SE practices. This modified framework is intended to enhance the process of transitioning from prototype development to a commercially viable product, with particular attention to stakeholder needs and expectations. It is expected that by employing this streamlined SE approach, companies will be able to develop their products efficiently, meet deadlines, and align the product with market requirements, all while effectively managing their limited resources.

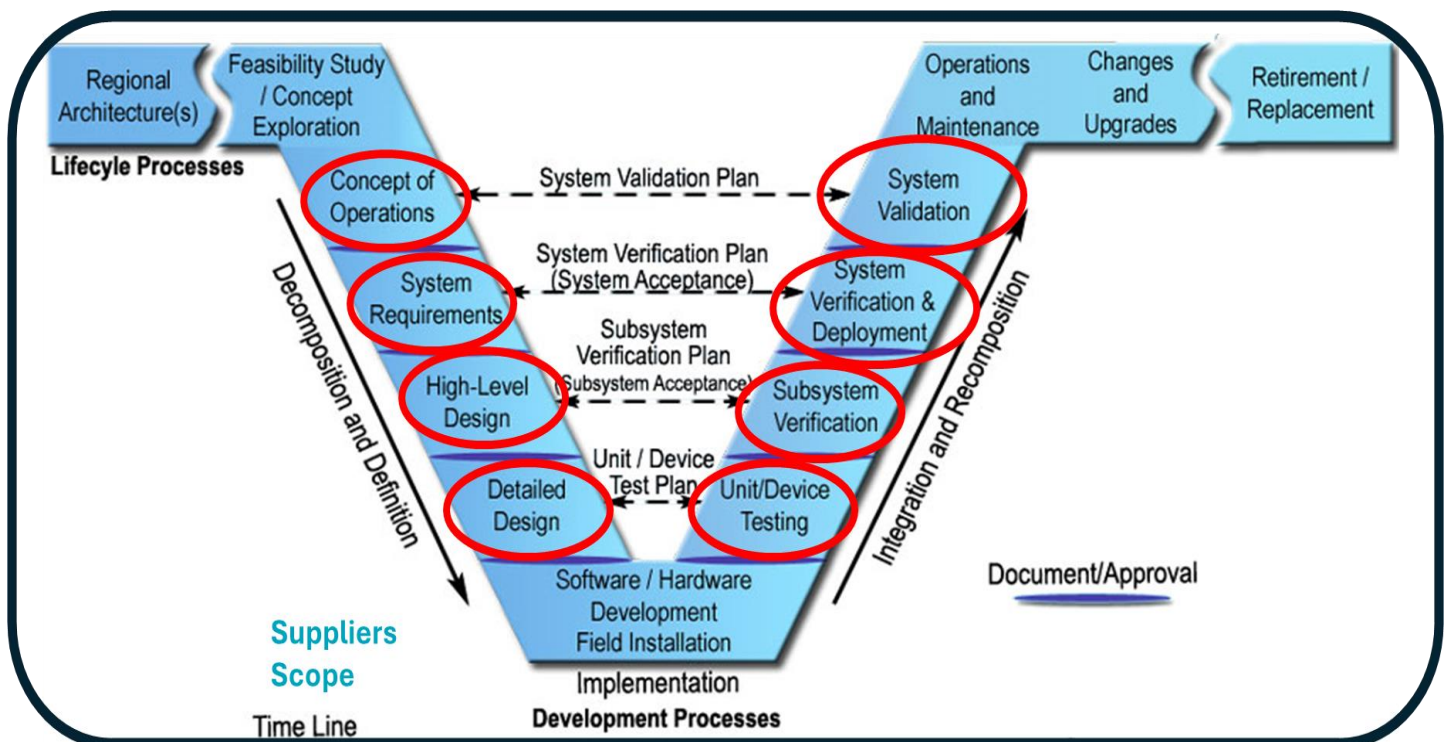
The article also discusses the challenges that will be faced during the research process, the conditions of the Master's degree, and the strategies that will be employed to address these challenges.

Appendix

The following figure shows the Infinity Fabrtkk Organization Charter.

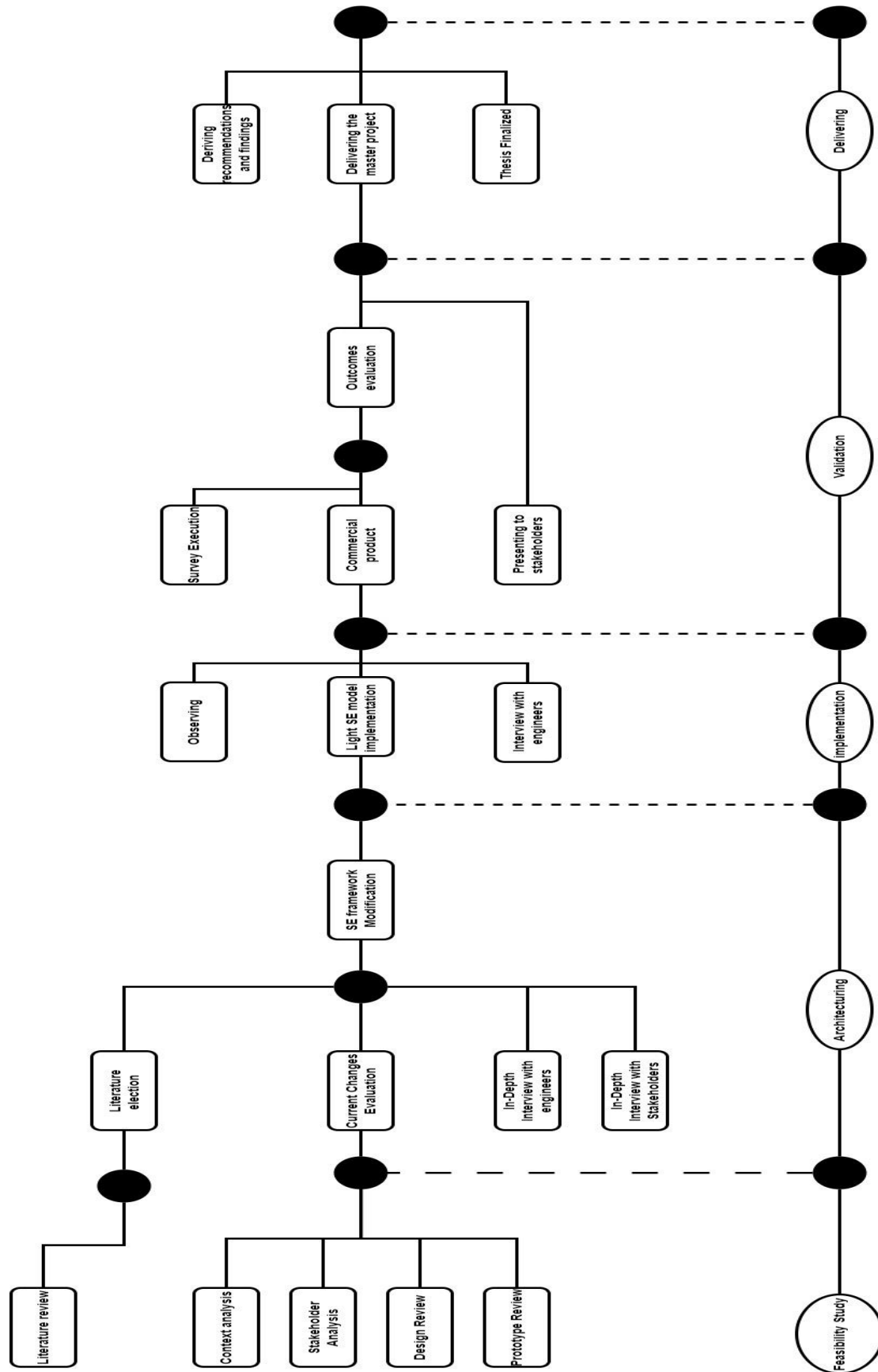


1. The stages that will evaluate the integration of systems engineering frameworks into its implementation process.



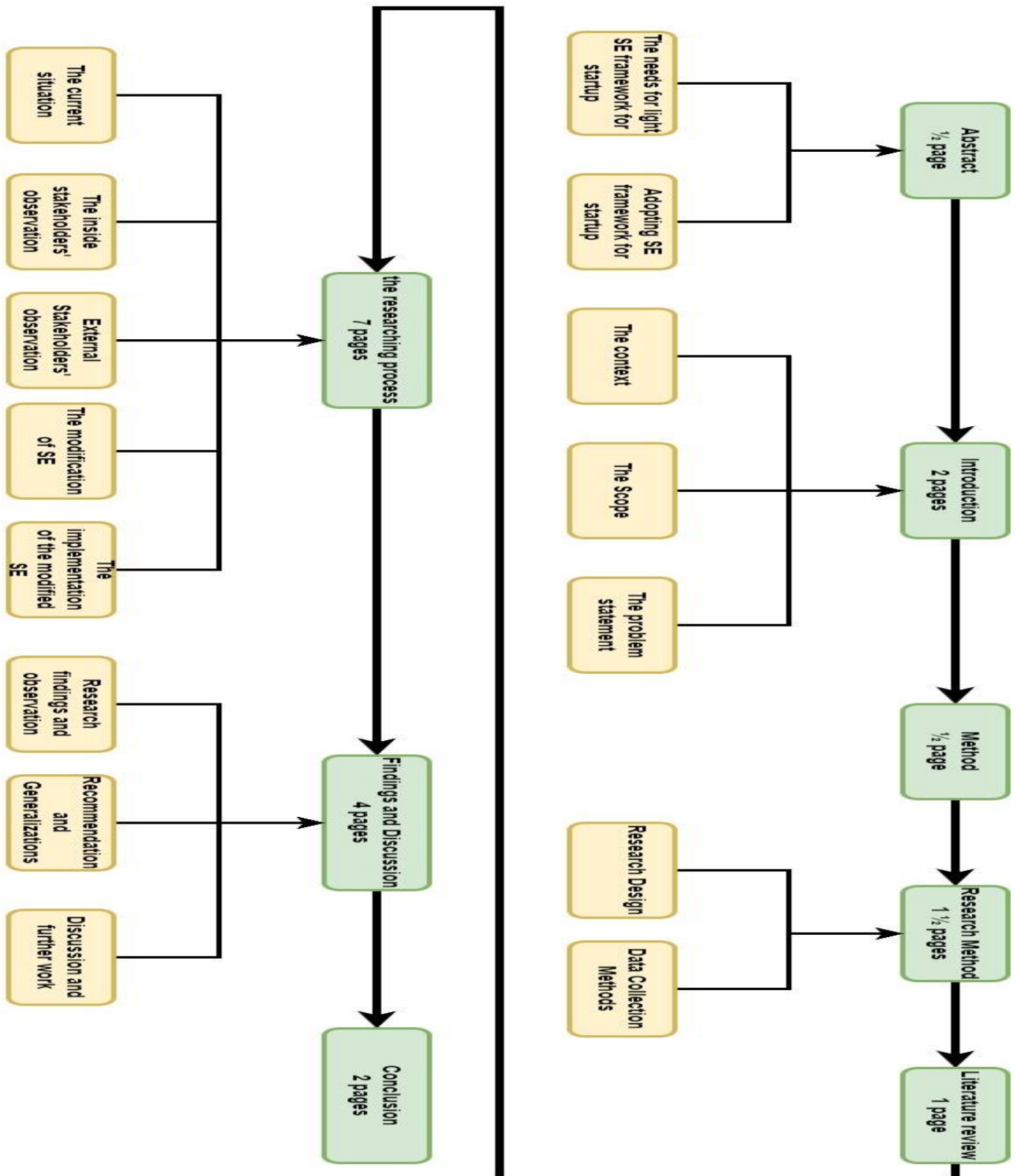
The Execution Plan

The following figure shows the PERT plan, which outlines how the project will be implemented.



Master Project Book Plan

The following figure shows the outline of the Book Plan, which shows the structure of the final article.



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